

Assignment

1. What is a file? Explain the need of a file in Python.

Ans:

A **file** is a named location on a storage device used to store data permanently. Unlike variables, which store data temporarily in memory, files retain data even after the program is closed.

Need of files in Python:

- To store data permanently.
- To read and write large amounts of data.
- To save program output for future use.
- To exchange data between different programs.
- To maintain records such as logs, reports, and user information.

2. State and explain the different modes of opening a file in Python.

Ans:

Mode	Description
r	Opens a file for reading . It gives an error if the file does not exist.
w	Opens a file for writing . Creates a new file if it doesn't exist and overwrites existing content.
a	Opens a file for appending . Adds new data at the end without deleting existing data. Creates the file if it doesn't exist.
x	Creates a new file . Raises an error if the file already exists.
r+	Opens a file for both reading and writing .
w+	Opens a file for reading and writing , but overwrites the existing content.
a+	Opens a file for reading and appending . New data is added at the end of the file.
rb	Opens a file in binary read mode.
wb	Opens a file in binary write mode.
ab	Opens a file in binary append mode.

3. State the difference between Write and Append mode in opening a file in Python.

Ans:

Write Mode (w)	Append Mode (a)
Overwrites the existing file content.	Adds new content to the end of the file.
Creates a new file if it does not exist.	Creates a new file if it does not exist.
Previous data is lost.	Previous data is preserved.
File pointer starts at the beginning of the file.	File pointer starts at the end of the file.

Example:

```
# Write Mode
file = open("data.txt", "w")
file.write("Hello")
file.close()
# Append Mode
file = open("data.txt", "a")
file.write("\nWelcome")
file.close()
```

After running both programs, the file will contain:

```
Hello
Welcome
```